

## Review 2 - Solutions

$$\textcircled{1} X \sim N(\theta, 4)$$

$$\text{prior } \theta \sim N(2, 2)$$

$$X = 2$$

posterior? MAP?

Bayes rule (CTS) case:

$$f(\theta|x) \propto \frac{f(x|\theta) \cdot f(\theta)}{\int f(x|\theta') \cdot f(\theta') d\theta'}$$

$\theta$  is mean of normal distribution  $\Rightarrow$  its domain is  $[-\infty, \infty]$

$$f(\theta|x) = \frac{\frac{1}{\sqrt{2\pi \cdot 4}} \exp(-(\theta-2)^2/4) \cdot \frac{1}{\sqrt{2\pi \cdot 2}} \exp(-(x-\theta)^2/2)}{\int \dots d\theta'}$$

$$= \text{const} \cdot \exp(-(\theta-2)^2/4) \cdot \exp(-(2-\theta)^2/2) =$$

$$= \text{const} \cdot \exp\left(-\frac{(\theta-2)^2(2+1)}{8}\right) = \text{const} \cdot \exp\left(-\frac{(\theta-2)^2}{2 \cdot 4/3}\right)$$

Since  $f(\theta|x)$  is a density function, it must be normal (which is also  $\text{const} \cdot \exp(-\frac{(\theta-\mu)^2}{2\sigma^2})$ )

$$\text{with } \mu = 2, \sigma^2 = 4/3$$

$$\text{So, posterior distribution is } \frac{1}{\sqrt{2\pi \cdot 4/3}} \cdot \exp\left(-\frac{(\theta-2)^2}{8/3}\right)$$

$$\boxed{\theta \sim N(2, \frac{4}{3})}$$

$\oplus$  same mean as prior, but smaller var - we become more confident!

! In Bayesian problems, make sure twice before computing integral in the denominator that you actually need to compute it! It might take lots of time and be unnecessary!

MAP estimate  $\hat{\theta} = \operatorname{argmax}_{\theta} f(\theta|x) =$   
 $= \operatorname{argmax}_{\theta} (c \cdot \exp(-\frac{(\theta-2)^2}{8/3})) = 2$

Since  $\text{At } \theta=2 \exp(-\frac{(\theta-2)^2}{8/3}) = 1$  otherwise it is  $< 1$ .

Also, gaussian density achieves its maximum at  $\theta = \mu$ .

[2] Linear regression model is  $\alpha + (x_i - \bar{x})\beta + \epsilon_i$

$\epsilon_i = N(0, \sigma^2)$   
 You can either re-derive MLE  $\hat{\alpha}, \hat{\beta}, \hat{\sigma}^2$  or use formulas from the textbook:

$$\hat{\alpha} = \bar{y}$$

$$\hat{\beta} = \frac{\sum_{i=1}^n y_i (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\sigma}^2 = \frac{1}{n} \cdot \sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta}(x_i - \bar{x}))^2$$

$$x_i = 32 \quad 23 \quad 23 \quad 30$$

$$y_i = 28 \quad 25 \quad 24 \quad 27$$

$$n = 4$$

$$\bar{x} = \frac{1}{4} (32 + 23 + 23 + 30) = \frac{42 + 62}{4} = 27$$

$$\bar{y} = \frac{1}{4} (28 + 25 + 24 + 27) = \frac{80 + 13 + 11}{4} = 26$$

$$x_i - \bar{x} = 5 \quad -4 \quad -4 \quad 3$$

$$\hat{\alpha} = 26$$

$$\hat{\beta} = \frac{28 \cdot 5 + 25 \cdot (-4) + 24 \cdot (-4) + 27 \cdot (3)}{25 + 2 \cdot 16 + 9} = \frac{140 - 100 - 96 + 81}{25 + 32 + 9} =$$

$$= \frac{40-15}{66} = \frac{25}{66} \approx \frac{24}{66} = \frac{4}{11} \approx \frac{1}{3} \quad (*)$$

$$\boxed{y = 26 + (x - 27) \cdot \frac{1}{3}} \quad \text{— regression line}$$

Thus estimate for  $x = 22$  is

$$y = 26 - \frac{5}{3} = \left( \frac{73}{3} \right)$$

$$Q^2: \text{ if } x_i = 23 \quad \hat{\alpha} + \hat{\beta}(x_i - \bar{x}) = 26 + (23 - 27) \frac{1}{3} = \left( \frac{74}{3} \right)$$

$$\text{ if } x_i = 32 \quad \text{— " —} = 26 + \frac{5}{3} = \left( \frac{83}{3} \right)$$

$$\text{ if } x_i = 30 \quad \text{— " —} = 26 + \frac{3}{3} = 27$$

$$Q^2 = \frac{1}{4} \left( \left( 28 - \frac{74}{3} \right)^2 + \left( 25 - \frac{83}{3} \right)^2 + \left( 24 - \frac{83}{3} \right)^2 + (27 - 27)^2 \right)$$

$$= \frac{1}{4} \cdot \left( \left( \frac{84-74}{3} \right)^2 + \left( \frac{75-83}{3} \right)^2 + \left( \frac{72-83}{3} \right)^2 + 0 \right) =$$

$$= \frac{100 + 64 + 121}{9.4} = \left( \frac{285}{36} \right)$$

(\*) your answers would be different without simplifying approximation (\*)

you can do reasonable roundings on the final.

if you are not sure what rounding is "reasonable", please ask during the test!

③ We would like to test the hypothesis that coin is fair:  $H_0: p_1 = \frac{1}{2} = P(\text{heads}); p_2 = \frac{1}{2} = P(\text{tail})$   
There are 2 options, head or tail  $\Rightarrow$  number of classes  $k=2$

$$q_{2-1} = \sum_{i=1}^2 \frac{(y_i - n \cdot \frac{1}{2})^2}{n \cdot \frac{1}{2}} = \frac{(85-100)^2}{100} + \frac{(115-100)^2}{100}$$

/see next page/



$$= \frac{2 \cdot 15^2}{100} = \frac{225.2}{100} = \textcircled{4.5}$$

Critical region  $\{Z \geq q_{k-1}\}$

$$\chi_{0.05}^2(2-1) = \chi_{0.05}^2(1) \approx 3.8$$

Since  $\{4.5 > 3.8\}$ , we reject  $H_0$  (coin is not fair)

p-value:  ~~$\chi^2(1)$~~   $\textcircled{3.8}$

$$P(Z \geq q_{k-1})$$

where  $Z \sim \chi^2(1)$

$$P(Z \geq q_{k-1}) = P(Z \geq 4.5) \approx 0.035$$

(between 0.05 and 0.025 in the table)

Since  $p\text{-value} < 5\%$ ,  
we reject  $H_0$

$$\textcircled{4} f(x, \theta) = \frac{1}{\theta} x^{(1-\theta)/\theta}$$

Likelihood function is

$$L(\theta) = \prod_{i=1}^n f(x_i, \theta) = \left(\frac{1}{\theta}\right)^n \prod_{i=1}^n x_i^{(1-\theta)/\theta} = \frac{(\prod x_i)^{(1-\theta)/\theta}}{\theta^n}$$

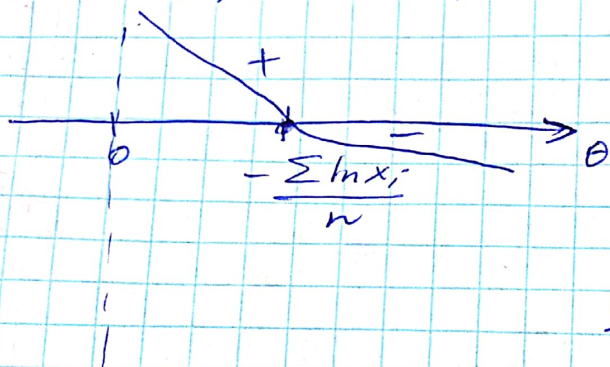
$$\ln L(\theta) = -n \ln \theta + \frac{1-\theta}{\theta} \left( \sum_{i=1}^n \ln x_i \right) \rightarrow \max ?$$

$$(\ln L(\theta))' = -\frac{n}{\theta} - \frac{1}{\theta^2} \left( \sum_{i=1}^n \ln x_i \right) = 0$$

$$\frac{(\theta n + \sum \ln x_i)}{\theta^2} = 0 \quad \theta = -\frac{\sum \ln x_i}{n} \quad \text{or } \theta = 0$$

Which one gives max?

$$x_i \in (0,1) \Rightarrow \ln x_i < 0 \Rightarrow \sum \ln x_i < 0$$



$$f(\theta) = \ln(L(\theta))' = \frac{-n\theta - \sum \ln x_i}{\theta^2}$$

$f(\theta) > 0$  for small  $\theta$

$f(\theta) < 0$  for large  $\theta$

$$\text{So, } \boxed{\hat{\theta} = -\frac{\sum \ln x_i}{n}}$$

$$b) \quad \mathbb{E} \hat{\theta} = -\frac{\sum_{i=1}^n \mathbb{E}(\ln x_i)}{n}$$

$$= -\frac{\sum_{i=1}^n \frac{1}{\theta} \mathbb{E}(\ln(x_i))}{n}$$

$$= -\frac{\sum_{i=1}^n \theta}{n} = \frac{n\theta}{n} = \theta$$

$\Rightarrow \hat{\theta}$  is an unbiased estimator

$$\mathbb{E} \ln X = \int_0^1 \underbrace{\frac{1}{\theta} t^{(1-\theta)/\theta}}_{v'} \underbrace{\ln t}_{u} dt =$$

$$= \left[ \frac{1}{\theta} t^{1/\theta} \ln t \right]_0^1 - \int_0^1 t^{1/\theta} \cdot \frac{1}{t} dt =$$

$$= 0 - \int_0^1 t^{1/\theta - 1} dt = \left[ \theta t^{1/\theta} \right]_0^1 = \theta$$

$$\left( \begin{array}{ll} v = t^{1/\theta} & v' = t^{1/\theta - 1} \cdot \frac{1}{\theta} \\ u = \ln t & u' = \frac{1}{t} \end{array} \right)$$